PROOF OF CONCEPT · PROGRESS UPDATE

Bullhorn Cross-Company Candidate Matching

Date: 15 July 2026 **Phase:** Foundation complete · search design proven · awaiting decisions

The secure connection to both companies' Bullhorn systems is built and working. Everything about the *search itself* is designed and proven feasible against the real data, but not yet built — we deliberately paused to surface what the data investigation found, because it changes what the matching can honestly promise.

1 What we've built so far

A secure, automated connection to both companies' live Bullhorn systems. This is the one piece of running software today, and it was the largest technical risk — so it is the right thing to have finished first.

- It logs in to each company automatically and handles the regional routing Bullhorn performs behind the scenes.
- It stays connected between requests, so we are not repeatedly logging in — Bullhorn limits how often that is allowed.
- It is strictly **read-only**: it only ever *reads* from Bullhorn and never writes anything back. These are the client's live production systems, so that guarantee matters.

Everything below this point — the search, the ranking, the privacy redaction — is **designed and tested for feasibility, but not yet built**. We know how to build it; we have not written it.

2 What we found in the data

We examined around 50,000 real candidate records across the two companies. These findings are the substance of this update.

Finding 1 — the two companies' data quality is very different

They cannot be treated as interchangeable:

	Company A	Company B
Candidates	~12,200	~38,300
Have skills recorded	80%	42%
Reachable by a skills search	76%	43%

> **EXAMPLE**
> If a recruiter searches Company B for a specific skill, more than half its candidates — roughly **22,000 people** — can never appear in the results. Not because they lack the skill, but because nobody ever recorded any skills for them.

Finding 2 — the "years of experience" field is unusable

Every candidate in both companies has their experience recorded as the number "1."

> **EXAMPLE**
> A superintendent with 30 years on site and a fresh graduate both show "1" in Bullhorn. It is a default value nobody filled in, so we cannot rank or filter by seniority using this field.

Finding 3 — we worked out how Bullhorn's search actually behaves

Bullhorn does not document its search rules, so we learned them by running real searches and checking the results. Two of the original assumptions were wrong, and in both cases the search would have silently returned *nothing*, with no error to warn us:

- **Skills can't be searched by name.** Searching for the word "Concrete" returns zero candidates. Each skill has a hidden reference number — "Concrete" is number 1000003 — and you must search by that. Done correctly, it returns 537 candidates.
- **A keyword search must be told where to look.** Searching for a loose word like "concrete" only works if you explicitly point Bullhorn at the right field; it will not search everywhere on its own.

We caught both before writing any code, and we now know the correct approach for each.

Finding 4 — the core design idea works

The plan is to treat some inputs as **strict requirements** (they remove unsuitable candidates) and others as **preferences** (they move the best matches to the top without removing anyone). We confirmed Bullhorn can do this.

> **EXAMPLE**
> Job titles are inconsistent — "Site Manager," "Construction Manager," and "Project Manager" can be the same role. If we treated the title as a strict requirement, we would delete good people who use a different title for the same job. As a preference, it ranks them higher without excluding anyone.

Finding 5 — we proved it end to end on real data

As part of the investigation we hand-ran one complete, realistic search against Company A — *"Project Managers who know Concrete and Piping."* It returned **76 genuinely relevant candidates**, with the strongest matches at the top. This was a manual test, not the finished product, but it shows the approach produces a sensible shortlist against the client's real database today.

3 Current limitations

- **Seniority cannot be used at all** — the experience field holds no real data.

- **Skills coverage is partial**, and much weaker on Company B, where 57% of candidates are unreachable by a skills search.
- **Job titles are not standardised** and can sometimes identify a specific person, so they need careful handling.
- **A candidate "do-not-share" flag** we expected to rely on is empty on both systems.
- **No test environment** — we have only the live production systems, though everything we do is strictly read-only.

The common thread: the information the matching most needs — real skills and real experience — largely does not exist in Bullhorn's structured fields. Agreeing how to handle that is the next decision point with the client.